

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

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2012 ACM Subject Classification Replace ccsdesc macro with valid one

Keywords and phrases Dummy keyword

Digital Object Identifier 10.4230/LIPIcs.CVIT.2016.23

Funding *Jane Open Access:* (Optional) author-specific funding acknowledgements

Joan R. Public: [funding]

Acknowledgements I want to thank ...

1 Introduction

ELPI is a dialect of λ PROLOG (see [14, 15, 7, 12]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 19, 8, 9]. Examples include the Hierarchy-Builder library-structuring tool [5] and Derive [17, 18, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI gained a static analysis for determinacy [10] to help users tame backtracking. ROCQ users are familiar with functional programming but not necessarily with logic programming and uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checkers identifies predicates that behave like functions, i.e., predicates that commit to their first solution and leave no *choice points* (places where backtracking could resume).

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantic depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI's implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [16] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard PROLOG abstract machines [20, 1], but it makes reasoning about the scope of *cut* difficult. To address that limitation we introduce a tree-based semantics in which the branches pruned by *cut* are explicit and we prove the two semantics equivalent. Using the

¹ Optional footnote, e.g. to mark corresponding author



```

Inductive P := IP of nat. Inductive D := ID of nat. Inductive V := IV of nat.

Inductive Tm :=
  | Tm_P of P
  | Tm_D of D
  | Tm_V of V
  | Tm_App of Tm & Tm.

Record Unif := {
  unify : Tm → Tm → Σ → option Σ;
  matching : Tm → Tm → Σ → option Σ;
}.

```

■ Figure 1 Tm types

■ Figure 2 Unif type

tree-based semantics we then show that if every rule of a predicate passes the determinacy analysis, the call to a deterministic predicate does not leave any choice points.

2 Common code: the language

Before going to the two semantics, we show the piece of data structure that are shared by the them. The smallest unit of code that we can use in the language is an atom. The atom inductive (see Type 1) is either a cut or a call. A call carries a term (see Figure 1). A term (Tm) is either a predicate, a datum, a variable or the binary application of two terms.

```

Inductive A := cut | call : Tm → A.
Record R := mkR { head : Tm; premises : list A }.
Record P := { rules : seq R; sig : sigT }.
Definition Σ := {fmap V → Tm}.
Definition bc : Unif → P →  $\mathcal{F}_v$  → Tm →
  Σ →  $\mathcal{F}_v$  * seq (Σ * seq A) :=

```

A rule (see Type 2) has a head of type term and a list of premises, that are a list of atoms. A program (see Type 3) is made by its rules and the signatures of predicates, **sigT** is a mapping from predicates to their signatures, signatures represent the definition of type from the simply typed lambda calculus, i.e. it is either a base type or an arrow for term application. We decorate arrows to know the mode (input or output) of the application argument.

A substitution (see Type 4) is a mapping from variables to terms. It is the output of a successful query. The empty substitution is noted ϵ .

The backchain function (bc, see Type 5) takes: a unificator U ; a program P ; a set fv of free variables ($\mathcal{F}_v$); a query q ; and the substitution σ . A backchain is performed when the current atom is a **call**. In this case, it start by recursively *dereferencing* the variables in σ , i.e. replacing all the variables in the query by their term mapping in σ . Then it takes the head of the term, which should be rigid, i.e. a predicate p , and look for the signature s of p in the program. Then, for each rule r in P , it unifies the head of r with q . Unification is possible thanks to the unificator U , whose type is shown in Figure 2. U explains how to unify terms up to standard unification (for output terms) or matching (for input terms). The function *unif* (resp. *matching*) unifies (resp. matches) the two terms under σ , it return an extension of σ if the two terms can be unified (resp. matched), and None otherwise. The result of a backchain is made by filtering the rules in P that can be used on the given query. In particular, for each rule filtered rules r , it returns a pair containing the extension of σ making the head of r and q unify and the list of premises of r .

```

f 1 2.    f 2 3.    r 0 0.    r 0 1.    r 0 2.
g X Z :- r X Y, f Y Z, !. % r1
g X Z :- f X Y, f Y Z.    % r2

```

Inductive tree :=
 | KO | OK | TA of A
 | Or of option tree & Σ & tree
 | And of tree & seq A & tree.

Figure 3 Small ELPI program example

Figure 4 Tree inductive

2.1 The cut operator

ELPI is a dialect of λ PROLOG with the hard cut operator and, in the language we have shown before, *cut* is an atom. The semantics of the cut operator adopted in the ELPI language corresponds to the *hard cut* operator of standard SWI-PROLOG. This operator has two primary purposes. First, it eliminates all alternatives that are created either simultaneously with, or after, the introduction of the cut into the execution state.

The ELPI program we will use as an example in the following sections is shown in Figure 3.

To illustrate the high-level description of *cut*, consider the program P shown in Figure 3 and the query $q = g \ 0 \ R$. Both rules for g can be used on the query q . Backchaining q in P from the empty (ϵ) substitution returns the list $[(\sigma_1, [r \ X \ Y, f \ Y \ Z, !]), (\sigma_2, [f \ X \ Y, f \ Y \ Z])]$ with σ_1 and σ_2 both equal to $\sigma_1 := \{X \mapsto 0, Z \mapsto R\}$. The two elements of the list are generated respectively from the rules r_1 and r_2 .

The next step of the interpretation corresponds to the backchaining of $r \ X \ Y$ from σ_1 . This returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [])]$ with $\sigma_3 := \sigma_1 + \{Y \mapsto 0\}$, $\sigma_4 := \sigma_1 + \{Y \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{Y \mapsto 2\}$. The three lists of premises are empty since all rules for r have no premises.

Backchaining $f \ Y \ Z$ in σ_3 gives no solution, therefore, by backtracking, we backchain $f \ Y \ Z$ in σ_4 . This gives a solution, namely $\sigma_5 := \sigma_4 + \{R \mapsto 2\}$.

We now reach the cut operator and can see its double side behavior. In the current execution state, we have still two unexplored alternatives: we can backtrack on the third solution for r that assign Y to 2 and on the rules r_2 . We can note, however, the first alternative have been generated after the cut introduction, and the second alternative have been generated at the same moment of the cut. Therefore both are cut away leaving no choice point. The final complete solution is: $\{X \mapsto 0, Z \mapsto R, Y \mapsto 1, R \mapsto 2\}$, where X , Y and Z can be ignored since they are the result internal unification.

3 Semantics intro

We propose two operational semantics for a logic program with cut. The two semantics are based on different syntaxes, the first syntax (called tree) exploits a tree-like structure and is ideal both to have a graphical view of its evolution while the tree is being interpreted and to prove lemmas over it. The second syntax, called elpi, is the ELPI's syntax and has the advantage of reducing the computational cost of cutting and backtracking alternatives by using shared pointers. We aim to prove the equivalence of the two semantics together with some interesting lemmas of the cut behavior.

4 Tree semantics

The tree inductive, shown in Figure 4. We distinguish 5 main cases: KO , OK , and are special meta-symbols representing, respectively, the basic failed and a successful trees. These

se metti $r1 = g \ A$
 $B :- f \ A \ B$. allora
 $g \ e \ f$ sono fun-
 zioni, e puoi spie-
 gare anche l'idea
 del detcheck qui

```

Fixpoint get_end s A :  $\Sigma$  * tree :=
  match A with
  | TA _ | KO | OK  $\Rightarrow$  (s, A)
  | Or None s1 B  $\Rightarrow$  get_end s1 B
  | Or (Some A) _ _  $\Rightarrow$  get_end s A
  | And A _ B  $\Rightarrow$ 
    let (s', pA) := get_end s A in
    if pA == OK then get_end s' B
    else (s', pA)
  end.

```

Definition get_subst s A := (get_end s A).1.
Definition path_end A := (get_end ϵ A).2.
Definition success A := path_end A == OK.
Definition failed A := path_end A == KO.
Definition path_atom A := if path_end A is TA _ then true else

■ **Figure 5** Definition of *get_end*

symbols are considered meta because they are internal intermediate symbols used to give structure to the tree.

The *TA* constructor (acronym for tree-atom) is the constructor of atoms in the tree. They are either the cut operator or a call, in these cases, the interpreter should evaluate the atom by either cutting the subtrees in the scope of the cut or backchaining the call in the program.

The two recursive cases of a tree are the *Or* and *And* nodes. The *Or* node $A \vee B_\sigma$ denotes a disjunction between two trees A and B . The first branch is optional, if absent it represents a dead tree, i.e. a tree that has been entirely explored. The second branch is annotated with a suspended substitution σ so that, upon backtracking to B , σ is used as the initial substitution for the execution of B .

The *And* node $A \wedge_{B_0} B$ represents a conjunction of two trees A and B . We call B_0 the reset point for B ; it is used to restore the state of B to its initial form if a backtracking operation occurs on A . An example of this behavior is shown in Section 4.2

A graphical representation of a tree is shown in Figure 6a. To make the graph more compact, the *And* and *Or* nodes are n-ary rather than binary, and the children of these nodes associate to the right. The *KO* node acts as the neutral elements in the *Or* list, while *OK* is the neutral element of the *And* list.

The interpretation of a tree is performed by two main routines: *step* and *next_alt* that traverse the tree depth-first, left-to-right. Then, then *run* inductive makes the transitive closure of step *step* and *next_alt*: it iterates the calls to its these auxiliary functions as much as needed. In Types 7–9 we give the types contrats of these symbols where $\mathcal{F}_v$ is a set of variable names.

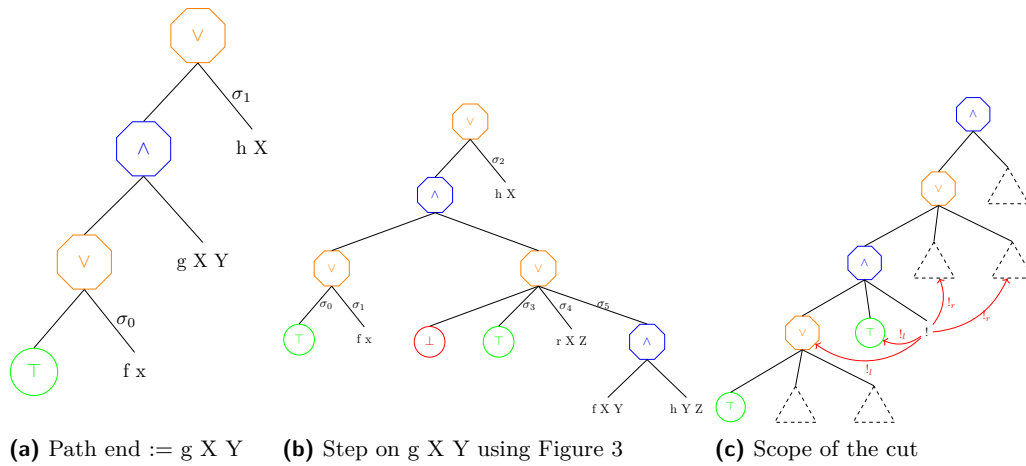
Inductive tag := Expanded | CutBrothers | Failed | Success. (6)

Definition step : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree}) :=$ (7)

Definition next_alt : $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ (8)

Inductive run (u:Unif) (p : $\mathbb{P}$): $\mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow \Sigma \rightarrow \text{option tree} \rightarrow \text{Prop} :=$ (9)

The tree interpreter, as in prolog, explores the tree in DFS strategy, to discover the substitution and the leaf of the tree that should be interpreted. The *get_end* routine, shown in Figure 5, accomplishes to this task. The *get_end* returns its inputs if the tree is a leaf. Otherwise, if the tree is a disjunction, the path continues on the left subtree, if it exists, otherwise it recursively retrieves the wanted piece of information in the rhs using the substitution stored in the *Or* branch: the current substitution when we cross the rhs of a *Or* is the one store in the *Or* node itself. In the case of a conjunction, if the to-be-explored leaf in the lhs is *OK*, then we look for the *get_end* in the rhs, otherwise we return the result of the lhs.



■ **Figure 6** Some tree representations

144 We derive the following two functions from *get_end*:

145 **Definition** $\text{get_subst } s \ A := (\text{get_end } s \ A).1.$ (1)

146 **Definition** $\text{path_end } A := (\text{get_end } \epsilon \ A).2.$ (2)

147 In Figure 6a the *path_end* of the tree is *g X Y*.

148 Below we define three special kinds of trees depending on their *path_end*.

149 **Definition** $\text{success } A := \text{path_end } A == \text{OK}.$ (3)

150 **Definition** $\text{failed } A := \text{path_end } A == \text{KO}.$ (4)

151 **Definition** $\text{path_atom } A := \text{if path_end } A \text{ is TA _ then true else false}.$ (5)

152 The latter definition identifies path ending in an atom.

153 4.1 The *step* procedure

154 The *step* procedure takes as input a program, a set of free variables (fv), a substitution, and
155 a tree, and returns an updated set of free variables, a *step_tag*, and an updated tree.

156 Free variables are those variables that appear in a tree; they are used and updated when
157 a backchaining operation takes place.

158 The *step_tag* (see Type 6) indicates the kind of an internal tree step: **CutBrothers** denotes
159 the interpretation of a superficial cut, i.e., a cut whose parent nodes are all *And*-nodes.
160 **Expanded** denotes the interpretation of non-superficial cuts or predicate calls. **Failure** and
161 **Success** are returned for, respectively, *failed* and *success* trees.

162 The step procedure is intended to interpretate atoms, that is, it transforms the tree iff its
163 *path_end* is an atom, otherwise, it returns the identity.

164 **Lemma** $\text{succ_step_iff } u \ p \ fv \ s \ A: \text{success } A \leftrightarrow \text{step } u \ p \ fv \ s \ A = (fv, \text{Success}, A).$ (1)

165 **Lemma** $\text{fail_step_iff } u \ p \ fv \ s \ A: \text{failed } A \leftrightarrow \text{step } u \ p \ fv \ s \ A = (fv, \text{Failed}, A).$ (2)

166 *Call step* The interpretation of a call *c* starts by calling the *bc* function on *c*. The output
167 list *l* is taken to represent build the new subtree. If *l* is empty then *KO* tree is returned,
168 otherwise the subtree is a right-skewed tree made of *n* inner *Or* nodes, where *n* is the length
169 of *l*. The root has *KO* as left child. The lhs of the other nodes is a right-skewed tree of *And*
170 nodes. The *And* nodes are again a right-skewed tree containing premises of the selected rule.

171 A step in the tree of Figure 6a makes a backchain operation over the query *g X Y* and, in
172 the program defined in Figure 3, the new tree would be the one in Figure 6b. We have put a

dire dei reset
point

red border around the new generated subtree. It is a disjunction of four subtrees: the first node is the *KO* node (by default), the second is *OK*, since r_1 has no premises, the third and the fourth contains the premises of respectively r_2 and r_3 .

Cut step The cut case is delicate, since interpreting a cut in a tree has three main impacts: at first the cut is replaced by the *OK* node, then some special subtrees, in the scope of the *Cut*, are cut away: in particular we need to soft-kill the left-siblings of the *Cut* and hard-kill the right-uncles of the the *Cut*.

► **Definition 1** (Left-siblings (resp. right-sibling)). *Given a node A , the left-siblings (resp. right-sibling) of A are the list of subtrees sharing the same parent of A and that appear on its left (resp. right).*

► **Definition 2** (Right-uncles). *Given a node A , the right-uncles of A are the list of right-sibling of the father of A .*

► **Definition 3** (Hard-kill, $!_r$). *Given a tree t , hard-kill replaces the given subtree with the *KO* node*

► **Definition 4** (Soft-kill, $!_l$). *Given a successful tree t , soft-kill replaces with *KO* all subtrees that are not part of the path in t leading to the *OK* node.*

An example of the impact of the cut is show in Figure 6c, the dashed triangles represent generic trees. The step routine interprets the cut since it is the node in its path-end: we pass through a and all trees on the left of the cut are successful. In the example we have 4 arrow tagged with the $!_l$ or $!_r$ symbols. The $!_l$ arrows go left and soft-kill the pointed subtree, it keeps *OK* nodes since they are part of the tree leading to the cut, and replaces the other subtrees with *KO*. The $!_r$ procedure replaces the nodes pointed by the arrows with *KO*.

4.2 The *next_alt* procedure

It is evident that the *step* alone is not sufficient to reproduce entirely the behavior of the full expected prolog interpreter. In particular, we need to bracktrack on failures. Moreover, in case of success, we should return a tree cleaned of its success, this is essential to, non deterministically, find the next solution of a given query. By Lemmas 1 and 2, we know that *step* returns the identity on successful and failed trees. In order to continue the computation on these particular trees, we need the *next_alt* procedure aiming to expecially work with failed and successful trees: and its implementation in Figure 7.

The *next_alt* procedure takes a boolean and a tree, clean it from failures or success and returns a new tree if this tree still contains a non explored path. The idea behind *next_alt* is to clean recursively every subtree in DFS order if its *path_end* is a failure. Moreover, if the boolean passed to *next_alt* is true, then it erases the first successful path in the tree.

The base cases of *next_alt* are immediate. The *Or* case is rather intuitive: if the lhs of the *Or* does not exist we look for the *next_alt* in the rhs. Otherwise, we look for the *next_alt* in the lhs, if this *next_alt* does not exists, we look for the *next_alt* in the rhs.

We want to spend few words about the *And* case, since the reset point B_0 for B plays an important role. The *next_alt* in an *And* tree should consider two cases: if the lhs succeeds, then the *next_alt* should be retrived in the rhs. If this alternative does not exists it means that the rhs has entirely been explored. We need to erase the success in the lhs and try to find if a non-explored alternative exists. If so, we return a new tree with the new lhs and the rhs is built from the reset point. *big_and* is a trivial function build a right-skewed tree of and nodes where the leaves are the atoms written in the reset point. We need to reuse the

```

Definition next_alt :  $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=
  \text{fix next\_alt } b \ A :=
    \text{match } A \text{ with}
    | KO \Rightarrow \text{None}
    | OK \Rightarrow \text{if } b \text{ then } \text{None} \text{ else } \text{Some OK}
    | TA _ \Rightarrow \text{Some } A
    | And A BO B \Rightarrow
      \text{let build\_BO } A := \text{And } A \text{ BO } (\text{big\_and } BO) \text{ in}
      \text{if success } A \text{ then}
        \text{match next\_alt } b \ B \text{ with}
        | None \Rightarrow \text{omap build\_BO } (\text{next\_alt true } A)
        | Some B' \Rightarrow \text{Some } (\text{And } A \text{ BO } B')
      \text{end}
      \text{else if failed } A \text{ then } \text{omap build\_BO } (\text{next\_alt false } A)
      \text{else } \text{Some } (\text{And } A \text{ BO } B)
    | Or None sB B \Rightarrow \text{omap } (\text{fun } x \Rightarrow \text{Or None sB } x) (\text{next\_alt } b \ B)
    | Or (Some A) sB B \Rightarrow
      \text{match next\_alt } b \ A \text{ with}
      | None \Rightarrow \text{omap } (\text{fun } x \Rightarrow \text{Or None sB } x) (\text{next\_alt false } B)
      | Some A' \Rightarrow \text{Some } (\text{Or } (\text{Some } A') \text{ sB } B)
    \text{end}
  \text{end.}$ 
```

■ **Figure 7** *next_alt* implementation

217 reset point since, the step procedure in *And* trees evaluates the rhs of a *And* tree if the lhs
 218 succeeds. This evaluation is dependent on the substitution in the lhs tree. Therefore, if we
 219 need to backtrack in the lhs, we need to reset the rhs.

220 Some interesting property of *next_alt* are shown below and allow to see how *next_alt*
 221 complements *step*.

222 **Lemma** *path_atom_next_alt_id* $b \ A$: $\text{path_atom } A \rightarrow \text{next_alt } b \ A = \text{Some } A$. (3)

223 **Lemma** *next_alt_failedF* $b \ A \ A'$: $\text{next_alt } b \ A = \text{Some } A' \rightarrow \text{failed } A' = \text{false}$. (4)

224 For example, in Figure 6b the step procedure has created a failed tree: its path-end ends
 225 in *KO*. The expected behavior of *next_alt* is to take this *KO* node and make it a This
 226 allows *step* to continue the exploration of the tree. In particular, the path-end of this new
 227 tree ends in *OK*. The step leaves the tree unchanged producing the new substitution. This
 228 solution however is not unique, we should be able to backtrack on this successful tree. To do
 229 so we can call *next_alt* and it will deadify the *OK* node allowing *step* to proceed on r X Z.

subst taken form
the or

230 4.3 The *run* inductive

231 TODO: start

232 DIRE CHE RUN TERMINA

233 In our implementation, *run* gives the guarantee that the program interpretation terminates,
 234 since it always return a substitution, but we can easily make it also accept diverging program
 235 by making *run* return an optional substitution

236 TODO: end

237 The inductive procedure *run* is modeled as a function: it takes as input a program, a
 238 set of free variables, an initial substitution σ_0 , and a tree t_0 , and returns a substitution σ_1
 239 together with an optional updated tree t_1 . The substitution σ_1 represents the most-general

unificator that makes the execution of the tree t_0 succeed starting from the initial substitution σ_0 , σ_1 is an extension of σ_0 . The tree t_1 is the updated tree containing the alternatives that have not yet been explored. If the tree contains no solution, then `None` is returned.

The procedure `run` is based on three main derivation rules, shown in Figure 8. If the `path_end` of the tree t is a success, the input substitution is returned and the input tree is cleaned of its successful path. If the `path_end` of the tree is an atom, then `step` is invoked to evaluate this atom, and `run` is recursively called on the new tree. Finally, if the `path_end` of the tree is a failure, `next_alt` is called to clear the failed path; if the resulting cleaned tree exists, `run` is recursively called on it.

$$\begin{array}{c}
\frac{\text{success } A \quad \text{get_subst } s_1 \ A = s_2 \quad (\text{next_alt true } A) = B}{\text{run fv } s_1 \ A \ s_2 \ B} \text{ RUN_DONE} \\
\\
\frac{\text{path_atom } A \quad \text{step u p fv0 } s_1 \ A = (\text{fv1, st, B}) \quad \text{run fv1 } s_1 \ B \ s_2 \ r}{\text{run fv0 } s_1 \ A \ s_2 \ r} \text{ RUN_STEP} \\
\\
\frac{\text{failed } A \quad \text{next_alt false } A = \text{Some } B \quad \text{run fv0 } s_1 \ B \ s_2 \ r}{\text{run fv0 } s_1 \ A \ s_2 \ r} \text{ RUN_FAIL}
\end{array}$$

■ **Figure 8** Rule system for `run`

248

249 **5 Elpi semantics**

We now want to introduce the elpi semantics. The interpreter we show reflects the interpreter of the ELPI language and is an operational semantics close to the one picked by Pusch in [16], in turn closely related to the one given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized the semantics in Isabelle/HOL together with some optimizations that are present in the Warren Abstract Machine [20, 1].

The inductive representing a state of the ELPI language is shown below.

```

Inductive alts := no_alt | more_alt of (Σ * goals) & alts
with goals := no_goals | more_goals of (A * alts) & goals .

```

An elpi state is an enhanced two-dimension list. The outermost list represents the list of alternatives in disjunction accompanate with the substitution that should be used to for their interpretation. The innermost list is a list of atom, representing a list of goals in conjunctions. These goals are decorated with a pointer to an elpi state, and are used to keep trace of the alternatives that should be kept when a cut is interpreted. We call these, special, alternatives the cut-to alternatives.

The idea of the ELPI interpreter is to receive a list of alternatives. The first alternative consists of a list of goals. Four cases must be taken into account; they are shown in Figure 9. In order to simplify goal retrieval, we split the head of the alternatives from the tail, so that it can be immediately matched in the inductive definition. Note that an empty list of alternatives represents, by definition, a failing elpi state. If the goal list is empty (STOPE), then we have, by definition, a success, and the input solution together with the list of alternatives is returned. If the goal list starts with a cut (CUTE), then the current alternatives are erased in favour of the cut alternatives, and a recursive call is made on the remaining goal list.

Definition $\text{stepE } \text{fv } t \text{ s } a \text{ gl} :=$
 $\text{let } (\text{fv}', \text{rs}) := \text{bc up fv } t \text{ s} \text{ in}$
 $\text{let } \text{rs_ca} := \text{save_alts } a \text{ gl } (\text{r2a rs}) \text{ in}$
 $(\text{fv}', \text{rs_ca}).$

Inductive $\text{nur} : \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{goals} \rightarrow \text{alts} \rightarrow \Sigma \rightarrow \text{alts} \rightarrow \text{Prop} :=$ (10)

$$\frac{}{\text{nur fv s } [::] \text{ a s a}} \text{STOPE}$$

$$\frac{\text{nur fv s gl ca } s_1 \text{ r}}{\text{nur fv s } [:: (\text{cut}, \text{ca}) \ \& \ \text{gl}] \text{ a } s_1 \text{ r}} \text{CUTE}$$

$$\frac{\text{stepE fv t s al gl} = (\text{fv}', [:: \text{b} \ \& \ \text{bs}]) \quad \text{nur fv' b.1 b.2 (bs++al) } s_1 \text{ r}}{\text{nur fv s } [:: (\text{call } t, \text{ca}) \ \& \ \text{gl}] \text{ al } s_1 \text{ r}} \text{CALLE}$$

$$\frac{\text{stepE fv t s al gl} = (\text{fv}', [::]) \quad \text{nur fv' } s_1 \text{ a al } s_2 \text{ r}}{\text{nur fv s } [:: (\text{call } t, \text{ca}) \ \& \ \text{gl}] [:: (s_1, \text{a}) \ \& \ \text{al}] s_2 \text{ r}} \text{FAILE}$$

■ **Figure 9** Rule system for *nur*

271 Finally, we must consider the case in which the goal list starts with a call. The call
 272 can either fail (FAILE) or succeed (CALLE). We distinguish the two cases by looking if the
 273 backchaining operation returns zero or more rules. We have wrapped this task in the *stepE*
 274 procedure, which also updates the goal and cut-alternative list. The fail case, is relatively
 275 easy: the first goal does not succeed, we need to take the head of the alternatives, and make
 276 it the new list of goals to be explored.

277 The case in which backchaining produces a non empty list, the *save_alts* routine is in
 278 charge of: taking the list of premises and add to each atom the the list of alternatives *a* as
 279 their new cut-alternatives, then it append the list of goals *gl* to each of these new lists.

280 6 Semantic equivalence

281 The equivalence between the two semantics is possible under two conditions: we need to
 282 work with “valid tree”, i.e. all of those trees that can be generated from a call. Secondly,
 283 we need to translate trees into elpi states. In the next to subsection we propose the two
 284 functions *valid_state* and *t2l*.

285 6.1 Valid trees

286 The inductive tree allows one to generate a large number of trees, some of which are not
 287 valid, in the sense that they cannot be produced starting from a given query. The class of
 288 valid trees is characterized by the function shown in Figure 10a.

289 While all base cases of tree are considered valid, we need to analyze carefully the cases
 290 for the *Or* and *And* constructors.

291 For the *Or* constructor, we distinguish two cases depending on whether the left-hand
 292 side (lhs) exists. If it does not exist, then the right-hand side (rhs) must be a valid tree.
 293 Otherwise, the lhs must itself be a valid tree, and the rhs is either the *KO* tree, since it may
 294 have been removed by the evaluation of a superficial cut in the lhs, or it has not yet been

```

Fixpoint t2l A s0 bt :=
  match A with
  | OK          => [:: (s0, [::])]
  | KO          => [::]
  | TA a        => [:: (s0, [:: (a, [::]) ])]
  | Or None s1 B => add_ca_deep bt (t2l B s1 [::])
  | Or (Some A) s1 B =>
    let lB := t2l B s1 [::] in
    let lA := t2l A s0 lB in
    add_ca_deep bt (lA ++ lB)
  | And A B0 B =>
    let lA := t2l A s0 bt in
    let lB0 := a2g B0 in
    let lA := add_deep bt lB0 lA in
    if lA is [:: (s0, gs) & al] then
      let al := map (catr lB0) al in
      let lB := t2l B s0 (al ++ bt) in
      map (catl gs) lB ++ al
    else [::]
  end.

Fixpoint valid_tree A :=
  match A with
  | TA _ | OK | KO => true
  | Or None _ B => valid_tree B
  | Or (Some A) _ B => valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B => valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

```

(a) Valid tree

(b) Tree to list

■ **Figure 10** Valid tree and Tree to list

explored. In the latter case, it is a `base_or` tree, namely the right-skewed tree formed by a disjunction of conjunctions that is generated after a successful backchain to a call.

For the `And` constructor, the lhs is required to be a valid tree. The shape of the rhs depends on whether the lhs is a success. If the lhs is not successful, then the rhs has never been explored: the procedures `step` and `next_alt` modify the rhs only when the lhs succeeds. In this case, the lhs must be the right-skewed tree containing conjunctions of premises atoms, the reset point B_0 ensure what shape the rhs should have. If the lhs is a success tree, then the rhs can be modified by `step` and `next_alt`, therefore it must be a valid tree.

6.2 From trees to lists

The translation of a tree to a list is shown in Figure 10b. It takes the tree to be translated, a substitution (called s), a list of alternatives, called bt . The substitution s tells what is the substitution of the alternative we are building and is updated when going to the rhs of the `Or` constructor. The bt list represents the future alternatives list that have the same depth of the current tree root. They are useful to know if they should be added to the current goal as cut alternatives.

An `OK` node represent a success in a tree, then the translation of this tree returns a singleton list with the couple $(s, [::])$: there is one alternative with 0 goals, i.e. a success in elpi by `STOPE`. The `KO` node represent a failure, and, therefore 0 alternatives. An atom is translated into a singleton with the atom as first goal and the empty cut-alternative list: we are not adding bt since they are the alternatives for the right-sibling, not the right-uncles, this means that if we have a is a cut, then it erases the alternatives bt .

In a disjunction, we are scendendo di un livello the distance from our backtracking list. This means that bt becomes the list of right-uncles of the two branches of the `Or` constructor. The compilation of the rhs is done independently of if the lhs exists: we transform the rhs

319 into an elpi state and passing the empty list of alternatives, since the rhs as no right-siblings.
 320 Then, thanks to `add_ca_deep`, we recursively add `bt` to every cut-alternative appearing in
 321 the translated goals. If the lhs exists, we translate it and pass the translation of the rhs as
 322 the list of right-siblings. Then we prepend the translated lhs to the translated rhs and add
 323 `bt` with `add_ca_deep`.

324 We compile the *And* case by translating its lhs to the list `lA` and reset point to the list
 325 `lB0`. Then, thanks to `add_deep`, we recursively append `lB0` to the alternatives born in `lA`,
 326 i.e. we leave unchanged the pointers to `bt`, if any, in `lA`. We finally need to treat cases
 327 whenever the list `lA` is empty or not. In the former case the empty list is returned, otherwise,
 328 we have a list $[(s_0, gs) \& al]$. By the semantics of the *And* in the *run* interpreter, the rhs
 329 of the *And* represent the sequent of goals to be executed after the first alternative in the
 330 lhs, it corresponds to the list of goals `gs`. The reset point `B0` is to be appended to the next
 331 alternatives in the lhs, that is the queue of `lA`, that is `al`.

alternative in
tree? rephrase

332 We note therefore that the compilation of `lB` is done by passing the alternatives `al + bt`
 333 since the alternatives in `al` are to be...

334 6.3 Equivalence theorems

335 We have now all the ingredients that ...

Lemma `tree_to_elpi`: $\forall u p s_0 t s_2 t',$
`let` `fv` := `vars_tree` `t` `|` `vars_sigma` `s_0` `in`
`valid_tree` `t` $\rightarrow$
 336 `run` `u` `p` `fv` `s_0` `t` `s_2` `t'` $\rightarrow$ (5)
 $\exists s_1 g a,$
`let`: `na` := `(t2l (odflt K0 t') s_0 [::])` `in`
`t2l` `t` `s_0` `[::]` = `(s_1, g) :: a` $\wedge$
`nur` `u` `p` `fv` `s_1` `g` `a` `s_2` `na`.

Lemma `elpi_to_tree`: $\forall u p fv s_1 g s_2 a na,$
`nur` `u` `p` `fv` `s_1` `g` `a` `s_2` `na` $\rightarrow$ (6)
 $\forall s_0 t, \text{valid_tree } t \rightarrow \text{t2l } t s_0 [::] = (s_1, g) :: a \rightarrow$
 $\exists t', \text{run } u p fv s_0 t s_2 t' \wedge \text{t2l (odflt K0 t')} s_0 [::] = na.$

338 The proof of Lemma 6 is based on the idea explained in [2, Section 3.3]. An ideal
 339 statement for this lemma would be: given a function `l2t` transforming an elpi state to a tree,
 340 we would have have that the the execution of an elpi state `e` is the same as executing `run` on
 341 the tree resulting from `l2t(e)`. However, it is difficult to retrieve the structure of an elpi state
 342 and create a tree from it. This is because, in an elpi state, we have no clear information
 343 about the scope of an atom inside the list and, therefore, no evident clue about where this
 344 atom should be place in the tree.

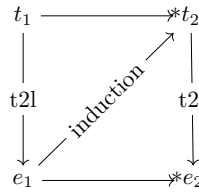
345 Our theorem states that, starting from a valid tree `t` which translates to a list of alternatives
 346 $(\sigma_1, g) :: a$. If we run in elpi the list of alternatives, then the execution of the tree `t` returns the
 347 same result as the execution in elpi. The proof is performed by induction on the derivations
 348 of the elpi execution. We have 4 derivations.

349 We have 4 case to analyse:

350 7 Case study: determinacy analysis

351 we mechanize the first order part of xxx.

352 snippet det, main thm, invariant det tree (valid tree prev section?)



■ **Figure 11** Induction scheme for Lemma 6

353 proof induction on exec, step/next alt preserving invariant proved by induction on the
354 tree.

355 with list semantics cut and next alt requires to express a link between the ca or next alts
356 and the current goal, which is non trivial without an intermediate data structure like the tree

357 8 Related work

358 prolog semantics, King lost
359 yves for the proof technique

360 9 Conclusion

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